

Executive Summary

Title:

Improving Stormwater Management in Parramatta using of Water Sensitive Urban Design (WSUD).

Key Information:

Stormwater runoff is a major cause of waterways pollution in the catchments of Sydney, and it provides two-thirds of dissolved contaminants present in Sydney Harbour (Sydney Water, 2023). Pollutants in the Parramatta River are also augmented by the impervious surfaces (roofs and carparks) in Parramatta, which carry heavy metals, nutrients, plastics and sediments.

In the present report, it is recommended that Water Sensitive Urban Design (WSUD) integration should be fortified in the Local Environmental Plan (LEP 2023) and Development Control Plan (DCP 2022) to reduce pollution, improve water reuse and strengthen ecological resilience. The Stormwater Catchment Asset Management Plan 2026-2035 emphasises the fact that more explicit design standards, regular maintenance plans and retrofitting of the old assets are needed. Enforcement of WSUD controls will help the council to achieve the NSW Water Quality Objectives and meet the community expectations by meeting the United Nations Sustainable Development Goals 11 (Sustainable Cities) and 13 (Climate Action).

The recommendations align with the Stormwater Catchment Asset Management Plan 2026-2035 that outlines the priorities of investment in stormwater infrastructure to achieve sustainability. This aligns with state priorities, reduces long-term maintenance costs and supports catchment scale ecological resilience.

Financial Implications:

- WSUD retrofits are estimated to cost \$2.5 million (2026- 2030).
- Funding through Stormwater Management Service Charge, Catchment Improvement Program of Sydney Water and NSW DPHI co-grants.
- Long-term savings expected to be achieved through reduced cost of sediment removal, flood damages and costs on repairing assets.

The lack of investment will raise maintenance costs and devalue the property as well as hinder the goal of Parramatta to be a swimmable river in 2025.

Recommendations:

That the Executive Group of City of Parramatta Council:

- Support a full WSUD policy which is compatible with the NSW Water Quality Objectives.
- Storm water offset contributions on retrofits.
- Biofiltration, wetlands and tree-pit systems should be prioritised in high-risk catchments.
- Have all WSUD assets maintained. Accept SOPA performance requirements of reuse of rainwater and deep-soil rehabilitation.

Visual Summary

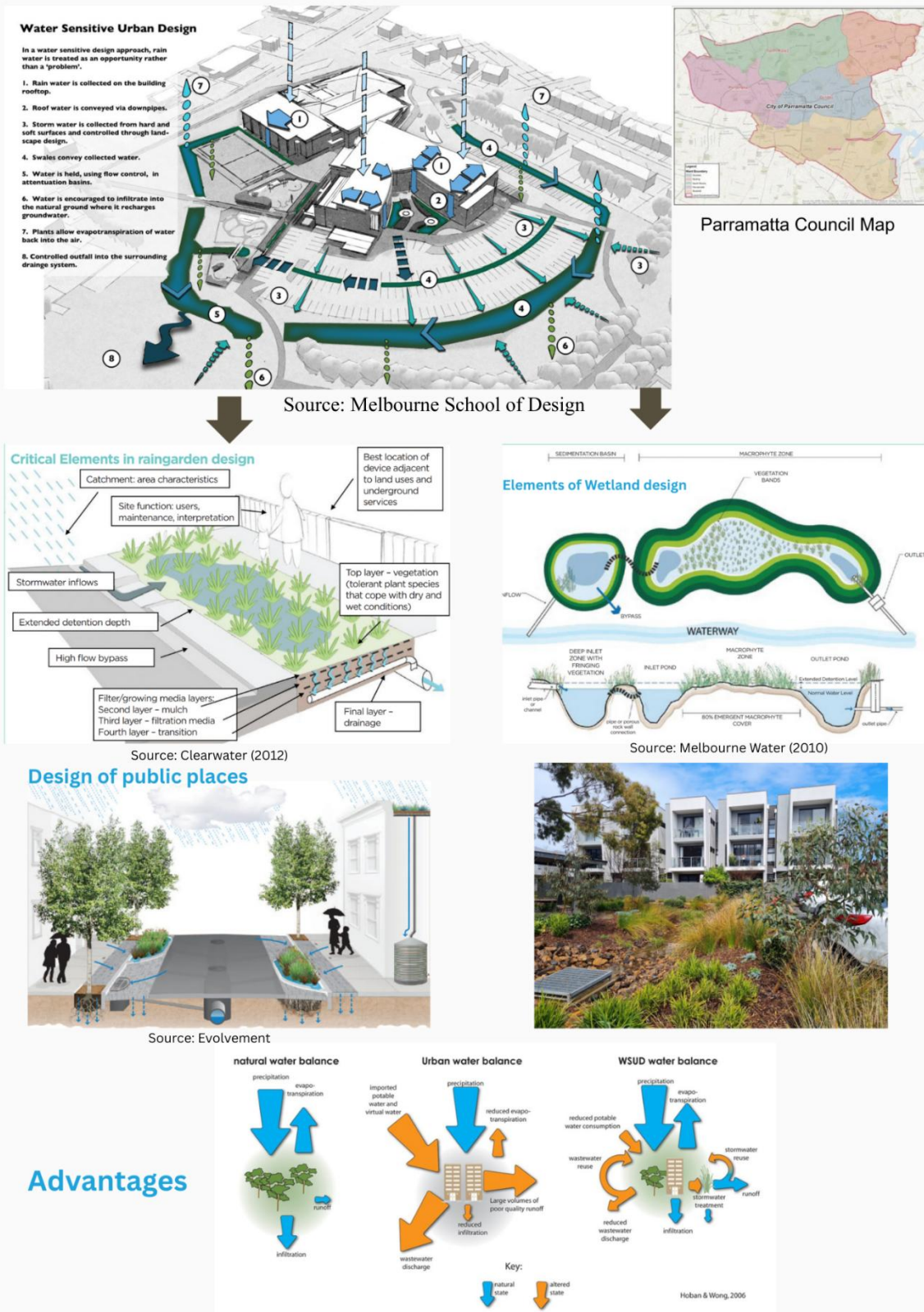


Figure 1: Water Sensitive Urban Design Model

Rationale

Alignment with Strategic Priorities

The introduction of the Parramatta River Masterplan and its corresponding waterway health projects are quite coordinated to the local and state strategic priorities. The council as a part of the Parramatta River Catchment Group makes its contribution towards improving the quality of the stormwater and creating new swimming areas in the river, which directly leads to the goal of the New South Wales Government to achieve a swimmable Parramatta River in the year 2025. This is also clearly supported in the Greater Sydney Commissions District Plans which requires councils to recognize waterways as essential green infrastructure and the promotion of water-sensitive urban design (WSUD) in new developments. By embracing the principles of WSUD and investing in the foreshore enhancements, the council will make sure that its work will be aligned with the overall vision of the Greater Sydney Region and District Plans: A City in its Landscape, which focuses on clean waterways and open spaces that are easy to access.

The strategy also promotes important goals provided in the Community Strategic Plan and connected corporate strategies. Indicatively, the Community Strategic Plan emphasizes the conservation and improvement of the natural resources like the Parramatta River, as well as promoting the growth of the recreational open space to the locals. Another area that requires more foreshore access and community activation is the North District Plan which includes the local government area. These priorities are directly met through the creation of new swimming sites and the enhancement of the health of the river as it will be turned into a useful public place. Recent example of the City of Ryde development of Putney Park into an urban beach is an excellent example of how local measures that are part of the Masterplan and District Plan can fulfil promises at the same time to make the river accessible and swimmable to residents.

Implications for Council Operations and Planning

The implementation of such a policy framework has significant implications on how the Council will plan, design and maintain urban infrastructure. First, internal procedures and guidelines need to be updated so that the water-sensitive urban design (WSUD) and the waterway health goals could be incorporated into the daily decision-making. An example of this is in the planning controls (LEP/DCP) which might be tightened to require on-site stormwater treatment, rainwater harvesting and permeable surfaces of new developments. The Stormwater Plan of the City of Parramatta now mandates that at least 1% of future civic upgrade area should have WSUD elements, which mainstreams water-sensitive design throughout the asset-delivery pipeline.

Secondly, the asset-management approaches and maintenance regimes should change. The construction of the large number of raingardens, gross pollutant traps, wetlands and infiltration systems requires budgeting and planning the maintenance of these items, such as regular sediment and litter removal, vegetation maintenance, and monitoring of performance. In the absence of proper maintenance, the performance of the WSUD assets reduces with time. This will then result in the higher staff capabilities in terms of special training in WSUD maintenance and water-quality monitoring, which may end up in new operational procedures or assigning special crews or contractors to care of water-sensitive assets, like the traditional maintenance work on drainage.

The other implication is related to the cross-departmental cooperation in the Council. The project is integrated, and it needs to be coordinated by involving the environmental services and strategic planning teams, capital works and community engagement teams to deliver the Masterplan.

It is also important to financial plan. Most Masterplan activities, including installation of new gross pollutant traps, wetlands, pipes used in stormwater to reduce overflows or developing river beaches and facilities are costly. These costs must be given precedence in the Delivery Program and outside funding should be obtained where possible.

Lastly, regarding the governance and partnership aspect, the implementation of this framework commits the Council to a good working relationship with the outside agencies and neighbouring councils. This Masterplan specifically supports a whole-of-government approach, whereby there are clear governance arrangements which are headed by Sydney Water.

Efficacy of the Policy Framework

The initial indicators are that the combined strategy provided in the policy framework shows optimistic results, but the ultimate success depends on the consistent, long-term work. As the Parramatta River Masterplan was promulgated in 2018, the quantifiable changes in water quality have been seen in parts of the river, which were accompanied by the opening of new swimming facilities on time. The Masterplan has also led to the creation of three more riverine swimming venues, including Bayview Park, Putney Beach, and a location at Rhodes and thus supplementing the existing Lake Parramatta and Harbourside Baths. The joint monitoring of these areas with the Sydney Water and council teams has shown that most of these sites are used more as recreational facilities, which can be explained by the fact that the pollution-reduction measures were implemented. According to Ian Wright of the Western Sydney University, the concerted efforts to date have made the waters swimmable, most of the time, and that the quality of the water has improved.

One of the factors that will influence the effectiveness of the policy is the establishment of Water Sensitive Urban Design (WSUD) and green cover in the catchment. Empirical studies and case studies indicate that, when properly designed WSUD characteristics, the urban runoff pollution can be significantly mitigated, and the peak volumes of the flow can be decreased.

The other measure of efficacy is the achievement of positive community outcomes and stakeholder buy-in. The preliminary statistics show that the society is positively receiving the new swimming sites; the population is busy using the new swimming sites and they are confident in the new monitoring systems. This increase in the level of public participation is a factor in independent success, which will increase the perceived usefulness of the river in the community and will contribute to the further support of investment.

Risks and Benefits of the Recommended Approach

The suggested proactive solution (the implementation of the Masterplan in its entirety and the increased integration of WSUD) has several benefits to our organization, along with some risks, which will require efficient management measures. The following is a list of the major advantages and risks and the mitigation strategies suggested on the risks:

Environmental Impact: The first advantage is the cleaning up and improvement of the riverine ecosystem, which results in the high quality of water and ecological stability. This will

improve the quality of aquatic habitats and biodiversity will be diversified in the Parramatta River catchment by reducing stormwater pollution and sewage overflows.

Community Health & Wellbeing: The availability of safe swimming and recreation in the river will be a new community facility to the locals. This has a direct transfer of health and social good whereby people can exercise, enjoy nature and later record better health outcomes in their mental health. It also deals with equity issues since it provides equitable access to recreational aquatic resources.

Strategic Compliance and Funding: The proactive implementation of the Masterplan puts the Council ahead of changing regulatory frameworks and initiative makes the Council strong with state-level institutions. Environmental standards in the state, such as water quality, urban cooling, and other such indicators, will become even more stringent. Operating now will reduce future compliance risks and be able to impact the establishment of standards instead of responding later.

Climate Resilience & Innovation: There are many Masterplan actions that result in co-benefits to climate change adaptation. WSUD elements like wetlands and canopied raingardens, serve both purposes like water purification and mitigation of urban heat islands, and carbon sequestration. A better stormwater management regime will increase the stormwater resilience to floods by reducing peak runoff rates, which will be instrumental in a future where extreme rainfall events will happen more often.

Key Risks and Mitigation:

Monetary and Resource Strain: A strong danger is the expense of the implementation. The upgrading of the infrastructure, the construction and maintenance of WSUD assets takes a lot of investment. There may be a strain on Council budgets and the capacity of the staff, particularly in the case of unforeseen costs (e.g., having to clean up a contaminated area discovered during the works).

Mitigation: We will address this through the mitigation of the implementation (spreading out projects over several financial years and ensuring that they are in line with our Asset Management Plan to prevent spikes in the expenditure). The aggressive seeking of external revenue will be maintained that comprises state grants, federal environment programs, developer contribution plans (Section 7.11 or 7.12 funds) aimed at water improvements. We shall also vent into the cost-sharing of neighbouring councils over shared projects (catchments cross boundaries) and joint ventures with the private sector on such matters as raingardens in new residential areas.

Maintenance and Performance Risks: It may not work properly or even cause nuisance (e.g. clogged bioretention basins may become breeding grounds of mosquitoes or will flood local streets) unless the WSUD installations and new facilities are well-maintained. This presents a danger to the environmental performance as well as Council reputation/service levels.

Mitigation: During project planning phase, include the maintenance requirement of the individual initiatives i.e. each new raingarden or gross pollutant tap must have a maintenance plan and budget. We might have to upgrade or recruit experts in maintenance. We will also have a monitoring regime (which may be community volunteers or adopt a raingarden programs). Through performance pointers (such as the number of pollutants captured, infiltration rates, etc), we will be able to arrange pre-failure rehabilitation of assets.

Expectations and Safety in Community: Public expectations are promoted when the river is publicized as being swimmable and when foreshore access is improved. This is in danger of community backlash in case e.g. of a high-profile pollution incident, or a swim site must be closed too often because of poor quality of water. The most important thing is to have the safety of people and any health-related incident (such as people falling ill because of being in contact with the water) would be a great setback to the program.

Mitigation: This will be controlled through strict water quality surveillance and open communication to the community. The multi-agency taskforce (including Sydney water and NSW health) will also supply real-time information and will send warnings at inappropriate times. Council will reinforce these warnings both physically and through the digital media. Also, by keeping on with the upstream pollution reduction efforts (such as stormwater harvesting and sewer repairs) we vigorously strive to ensure that the likelihood of lapses in water quality is reduced. Essentially, the program has cautions and consistent improvement incorporated to protect the community.

Inter-agency Dependence: We do not have complete control over the success of our plan. It is based on other councils (state agencies) playing their roles. As an example, the Sydney Water needs to minimize the wastewater overflow, and the EPA needs to impose the industrial pollution. In case any of our partners underperform or the priorities of the state's changes, our Council might be left to handle a problem with very few means (e.g. we cannot upgrade the sewers of Sydney Water directly).

Mitigation: To lessen this, we will participate actively in the governance forums (PRCG technical committees, District Water Quality groups) to hold the partners accountable and keep in the track of their developments. It is also possible to motivate partners to remain committed by building strong relationships and showing them local successes. Where we can anticipate lapses (such as a hold-up in a sewer upgrade program in our locality), then we can lobby our local MPs and via the media, using the support of the community to the health of the river to get things going. Within ourselves, we make sure that we do not rest on our laurels on the things we can do.

Alternatives Considered (Including Status Quo)

When developing our recommendation, we considered other possible solutions such as the status quo (minimal intervention) and other possible strategies. The list of these options and why the one suggested is better than the other is provided below:

Status Quo / Business-as-Usual: This would imply that Council just applies current, existing water management practices e.g. meeting the current requirements of the law on stormwater on new developments, having existing drainage systems, however not actively extending WSUD or implementing new projects based on the plans of the Masterplan. The advantage of this strategy is in the saving of costs in the short term and the lessening of the load on resources. We would save the direct cost of upgrades and would be able to channel employees to other urgent matters. The risks of the status quo are, however, high. Unless further measures are taken, the quality of water in the Parramatta River is expected to deteriorate with the further urbanization. We would surely not achieve the objective of the region to have a swimmable river (the socially and environmentally beneficial results would be lost), and we may also be out of step with the state planning requirements that require councils to achieve better water outcomes. In the long run, failure to do anything new might turn out to be expensive.

Partial or Piecemeal Implementation: The other option that was considered was a middle-ground approach, which, however, only select elements of the policy framework were to be

implemented. One could consider, as an example, investing in community education and enforcement (which reduces litter and pollution) but does not invest in extensive new infrastructure or the other way round, creating a couple of showcases WSUD projects, but not the entire catchment-wide program. It would be envisioned to bring about some improvements at reduced cost. This approach too failed after analysis. The Masterplan and associated policies are very conscious of being holistic; the various activities (education, infrastructure, regulation, naturalisation) are intended to support each other. When we take only the one view, the total effect is much lessened. An example is where we may construct new wetlands, but the enforcement against the illegal discharges or behaviour of communities may not improve, thus these new wetlands may contain high load of pollution and fail to perform.

Alternative Focus (Grey Infrastructure Only): We also thought about the possibility of replacing a green/WSUD approach with a more conventional one known as the grey infrastructure one. That would involve the use of engineered solutions such as large-scale water treatment systems or end-of-pipe filtration caustic and increasing the size of conventional drainage pipes to accommodate floods, rather than small WSUD characteristics. It would be rationalised by the potential to meet the water quality objectives by using centralised facilities (such as a new regional stormwater treatment plant) and controlling the risk of flooding using hard infrastructure. Although these strategies may be effective in some situations, they were not popular in our case. Constructing stormwater water treatment plant would be extremely costly and probably impractical due to a lack of land and cost in our urban town. It also does not provide the co-benefits of WSUD like greening and beautification of the community and introduces single points of failure (when a plant fails, the water quality drops). Further, this solution does not fit the principles of the integrated system of water cycle management of the State Government, and we would not have the opportunity to satisfy the expectations of the planning policies regarding the green infrastructure.

Context

The WSUD approach of Parramatta is one of the develop trends in urban water management, and several cities in the region and other parts of the world are seeking such solutions. The existence of at least three other councils gives us some perspective and some lessons, by comparing our efforts:

City of Melbourne (Australia): Melbourne is a leader in WSUD and combined water management. It was envisaged in the city as a Catchment strategy of the Total Watermark since 2008 that aimed at achieving ambitious goals of reducing stormwater pollution and use of alternative water. The Melbourne City Council had spent enormously with annually allocating about A\$2 million to stormwater quality initiatives solely installing more than 200 raingardens, permeable pavements and stormwater harvesting systems in parks (City of Melbourne, 2014). This has seen Melbourne cut water pollution flowing into its waterways by 16 percent (as of the early 2010s) and been noted to be leadership in its development of a water sensitive city. WSUD has been integrated in the planning regulations of Melbourne such that any new development should comply with the best practice stormwater treatment objectives (City of Melbourne, 2014).

Auckland (New Zealand): Auckland Council officially introduced the so-called Water Sensitive Design (WSD) as the most appropriate way of managing stormwater over the region. Another similar guideline is their 2015 GD04 guideline on Water Sensitive Design on Stormwater, which is comparable to WSUD and is now included in the Unitary Plan of Auckland (equivalent to a planning scheme). This implies that new developments in Auckland need to have on-site stormwater control such as rain gardens, green roof, or detention pond, unless non-practical, and large-scale greenfield development require an integrated stormwater management plan. Another important feature of Auckland's approach is that they

acknowledge Māori cultural values of water, hence WSD there as well is the naturalization of streams and preservation of riparian vegetation. (Auckland Council, 2015)

Vancouver (Canada): Vancouver has initiated a massive green infrastructure initiative in an effort called the Rain City Strategy. This strategy was adopted in 2019, and its overall headline is to ensure that 90 per cent of the annual rainfall in Vancouver is captured and treated on-site using green rainwater infrastructure by 2050 (City of Vancouver, 2019). It is among the most aggressive targets in the world indicating the intentions of Vancouver to essentially remove polluted runoff into its waterways. To do it, Vancouver is introducing 46 action programs such as the use of street-side infiltration swales, rain gardens in each neighbourhood, green roofs requirements and even permeable alleyways. They will manage 40 percent of impervious spaces in the city, by 2025 they expect to manage rainwater (City of Vancouver, 2019). The current number of green rainwater infrastructures on ground (bioswales, retention ponds, etc.) in Vancouver is more than 240, and it is intended to increase the amount of rainfall addressed by green infrastructure (24 mm to 48 mm of rainfall per day) (City of Vancouver, 2019).



Melbourne



Auckland



Vancouver

Figure 2: Water Sensitive Urban Design

Consultation

Internally, consultations with the Civil Infrastructure and Stormwater Operations team (manage drainage assets), Strategic Planning unit (alignment with planning controls and the Local Environment Plan/DCP) and the Parks and Environment team (opportunities of green infrastructure in open spaces) of the City of Parramatta Council and the Finance Management staff (insight on budgeting and asset planning).

External important stakeholders are:

Parramatta River Catchment Group (PRCG): PRCG as a regional group of councils and agencies offered strategic context of catchment-wide activities and objectives of the Parramatta River Masterplan. Their comments justified the Council initiatives to relate to the wider purpose of a swimmable river and continuing programs such as litter prevention and stormwater compliance programs.

Sydney Water: Since this organisation is the sewerage management authority as well as parts of the larger drainage infrastructure, Sydney Water was consulted with as to possible partnerships e.g. exploring the idea of stormwater harvesting or using monitoring data of Sydney Water on water quality. The local stormwater is not usually the business of Sydney Water, but they are experts in the areas of water recycling and network management, which are useful in integrated project work.

Government Departments: Particularly, the Department of Planning and Environment (Water and Metropolitan Units), and the NSW Environment Protection Authority. Consultations with these organizations made sure that the plans of Council do not conflict with any state policies and are able to access technical advice or funds.

Community and Environmental Groups: This internal report was not officially surveyed with these community and environmental groups, however, we used recent community feedback (including the Community Strategic Plan process and PRCG community surveys) that has repeatedly demanded cleaner waterways and increased green areas. Other organizations such as the Parramatta Climate Action Network and local Riverkeeper volunteers were able to give anecdotal information regarding the levels of popular support of raingardens and creek restoration.

Other agencies: There were informal contacts with other councils (City of Ryde, Cumberland City) and agencies such as the Sydney Olympic Park Authority (which has promoted WSUD policies). Such discussions provided lessons learned and the possibility to collaborate on projects, as the water problems have inter council borders (e.g. a wetland at Olympic Park can treat water that will be used by the entire river of Parramatta).

Next Steps and Collaboration: The Council will create a multi-disciplinary working group to facilitate the WSUD program as per the recommendation of the present report to ensure that all the sections (planning, assets, environment, finance) continue to feed into the program. At the exterior, Council will enhance partnerships using formal means. One of the subsequent steps is to consult the community on projects of WSUD. Also, the joint workshops will be organized with the state government professionals (e.g., the officers representing Water NSW or EPA may be invited to inform Council about the recent water quality standards). This collaborative model will have the effect that recommendations implementation will be inclusive, achievable and supported by the overall capacity of all the stakeholders involved in the sustainable water future of Parramatta.

Recommendations

Embed WSUD within Council Infrastructure: Change Council project planning checklists by setting the features of Water Sensitive Urban Design to be mandatory in all new or upgraded infrastructure across the Council. As an illustration, incorporate rain gardens or permeable pavements when re-constructing roads, tree pit bioretention to streetscapes and wetlands or swales to park redevelopments. Target at least 1 percentage of the area of a project to stormwater treatment landscaping (City of Parramatta Council, 2023; Water by Design, 2014). This will systematically increase green infrastructure run by the Council and enhance quality of storm water in the city.

Improve Development Controls and Compliance: Strengthen the City of Parramatta DCP by establishing definite numerical stormwater quality targets (in line with NSW best practice: e.g. 85% reduction of TSS, 65% reduction of TP, 45% reduction of TN load of new developments) (NSW Government, 2022). With WSUD as a routine development requirement, new development will not enhance pollution and flooding (Parramatta River Catchment Group, 2018).

WSUD Retrofit Program: Initiate a funded program to retrofit stormwater hotspots of high priority using WSUD interventions. Use the internal WSUD opportunity map of Council to

determine where biofiltration basins can be retrofitted like large car parks, drainage reserves or civic plazas, infiltration trenches, or rainwater harvesting systems. Introduce at least 2-3 pilot retrofit projects within the next 2 years to demonstrate benefits (e.g. roadside raingarden cluster in the CBD, wetland in a suburban park) (City of Parramatta Council, 2023; Melbourne Water, 2021). Determine the quality of water gains of these pilots to create the argument to expand the implementation. Retrofit should be combined with planned renewals of assets wherever feasible to maximize costs.

Enhance Gross Pollutant tap and litter management: Intensify the amount and effectiveness of gross pollutant tap (GPT) installations on the stormwater outlets of the Council with emphasis on drains discharging to the Parramatta River and its tributaries (Parramatta River Catchment Group, 2018; Sydney Water, 2023).

Stormwater Harvesting and Reuse: Find possibilities of local stormwater harvesting plans that will reduce overland flow and create a non-drinkable water - supply. As an example, explore the possibility of developing a stormwater harvesting system within the Parramatta CBD (as was already examined in Council plans) to harness and treat stormwater to irrigate parks and trees along the streets. The use of stormwater is an increase in stormwater reuse in line with the NSW urban water security aspirations and the conversion of a waste product into a resource (Infrastructure NSW, 2022; City of Parramatta Council, 2023).

Monitoring, Evaluation, and Reporting: Adopt an effective monitoring program to monitor the progress of stormwater management. This involves the monitoring of water quality at strategic creek locations (with PRCG and Beach watch where applicable) to assess enhancements over the period; asset management and monitoring of new WSUD installations (to ensure that they achieve the design treatment effectiveness); and the occasional audit of the compliance of development with WSUD (compulsory) (NSW EPA, 2022; PRCG, 2018).

Advocacy and Collaboration: Sustain advocacy on supportive policies and funding on the regional and state level. These involve sharing the experience of City of Parramatta in the creation of the Sydney Harbour Coastal Management Program as well as lobbying the government to fund large-scale WSUD over the long term (e.g. Urban Rivers grants, Green Infrastructure Fund) (Commonwealth of Australia, 2023; City of Parramatta Council, 2023).

Author Details

Md Fahim Rahman,

Environmental Planning Analyst (Intern),

City of Parramatta Council.



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Appendix A: Presentation Slide



Improving Stormwater Management in Parramatta Using WSUD

Md Fahim Rahman,
City of Parramatta Council

Why This Matters?

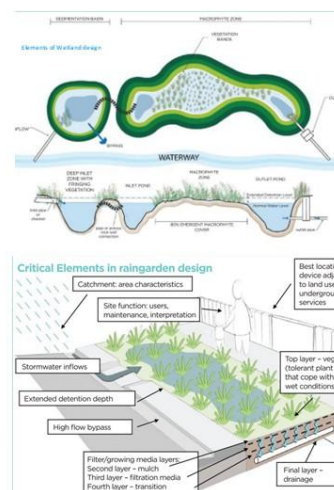
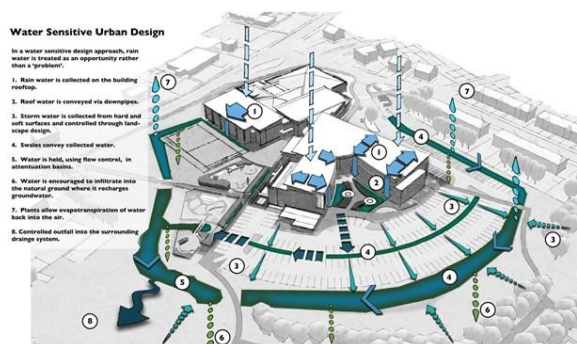
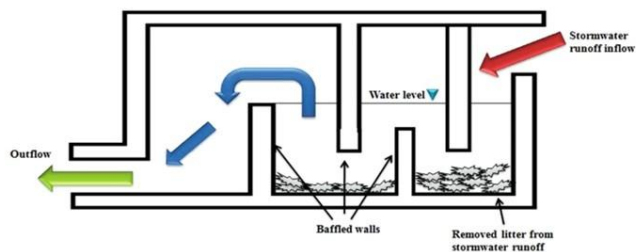
Stormwater runoff = major pollution source

- Provides ~ $\frac{2}{3}$ of dissolved contaminants in Sydney Harbour (Sydney Water, 2023)
- Impervious surfaces in Parramatta carry metals, nutrients, plastics, sediments
- Community goal: Swimmable Parramatta River by 2025 (City of Parramatta, 2023)



WSUD

- Biofiltration systems (raingardens, tree pits)
- Wetlands and swales
- Permeable surfaces
- Rainwater harvesting and reuse



Source: Melbourne Water (2010)
Clearwater (2012)
The project Moor Park Development

Policy Drivers & Planning Context

Frameworks that mandate WSUD

- Parramatta River Masterplan (PRCG, 2018)
- NSW Water Quality Objectives
- LEP 2023 & DCP 2022 embed WSUD controls
- Aligns with UN SDG 11 (Sustainable Cities) and SDG 13 (Climate Action)



Key Recommendations	Budget & Funding	Investment Area	Cost Over 4 Years
Strengthen WSUD adoption across Council • Embed WSUD in all capital works • Strengthen DCP stormwater standards • Prioritise biofilters, wetlands, tree pits • Stormwater offsets for retrofits	\$2.5M WSUD retrofits (2026-2030) Funding from: • Stormwater Service Charge • Sydney Water Catchment Improvement Program • NSW co-grant partnerships	Capital Works WSUD Integration	\$1.2M
		WSUD Retrofit Pilots	\$0.6M
		WSUD Maintenance Program	\$0.4M
		Compliance & Policy Support	\$0.3M
		Total	\$2.5M

Risks & Mitigation

Challenges

- Maintenance demand
- Potential community expectation gap
- Agency coordination risks

Mitigation

- Scheduled WSUD maintenance plan
- Clear communication on water safety
- Cross-agency governance with PRCG & Sydney Water

Benefits



Cleaner waterways & biodiversity uplift



Urban heat reduction & shade



Flood risk reduction



Higher property and public amenity value



Progress toward swimmable river goal

